



Figure 6: AWS WAF



Figure 7: AWS Certificate Manager

your Web app from common exploits that consume excessive resources, compromise security or affect application availability.

Certificate Manager: This helps in procuring SSL/TSL certificates for your AWS.

On-premise security

You can physically locate your hardware resources on-premise, and seeing the green LED flicker can lead you to assume that everything is alright. But once your resources are connected to the Internet, they are no more private. You can monitor the health of your server, but who can ever know the exact location of data once it's transferred from the source?

Even physical security of your hardware may be at risk if the intentions of those safeguarding your servers are not bona fide—perhaps because you delayed their payment for the past month or you are not paying them as per their expectations. You can always implant a CCTV camera, but this will only give you footage and not security.

Although it has been a major cause of debate between vendors and customers, cloud providers usually do not charge you for the maintenance of their hardware.

Public clouds increase security

Nowadays, cloud security is a hugely debated topic. All the cloud service providers guarantee maximum security. They provide services in three models—Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS). The main formula for successful security in any model provided by cloud service providers is 'restricted and required' access.

Even if the railway platform is for public use, there are certain rules that need to be followed before you step on to

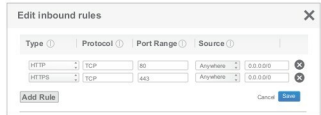


Figure 8: Security groups

the platform, otherwise you will get into trouble. The same way, these cloud service providers force their customers to impose certain security rules for public access. Some of these rules are listed below.

- 1. Security groups:** These are the groups that define the firewalls for a system. That system may be an instance, a database, a group of instances or even as large as your entire virtual cloud. These security groups have their own set of boundaries, namely, inbound and outbound range. You can add/delete/edit the rules defined inside these boundaries. You have the luxury to set your rules—from giving access only to an SSH user to Port 22 from some specific machine, to allowing access to the entire world.
- 2. Network access control lists (ACL):** This is an optional layer provided by some of the cloud service providers these days. In AWS, this acts as a firewall for controlling traffic to one or more subnets.
- 3. Key-pair access to VMs:** Key-pair is basically known as key cryptography. It is used to encrypt and decrypt the login information. The public key is used to encrypt the login data such as passwords, and then the private key is used to decrypt the data. This pair of public and private keys is known as a key-pair and the mechanism is known as key-pair cryptography. Many cloud providers use this key-pair to give their customers access to their virtual machines.
- 4. Identity and access management (IAM):** This allows organisations to enforce role based controls on what users can do in an AWS environment (IAM integrates with the users' existing Active Directory, or other authentication platforms).
- 5. Virtual private cloud (VPC):** This is like your data centre inside the cloud's data centre. You have control over configuring your own route table, access control list (ACL), Internet gateway, and subnet setting. Apart from these, there are some server level security systems like *ipconfig*, *sysconfig*, etc.

Possible threats and controls

In today's world, we are the greatest threat to our own data. I would like to divide these security threats into two broad categories.

Physical security threats

Physical security is the biggest reason why organisations